

Compliance in Databases: A Study of Structural Policies and Query Optimization

Observations

1. White-Box Implementation

- For execution time, both the View and Rewrite approaches outperform Pure RLS for the majority of queries. Query Rewrite generally achieves the best performance across the workload.
- In most queries, the Rewrite approach performs faster than both the View and Pure RLS approaches, indicating that query rewriting can expose optimization opportunities that are not utilized by the native RLS implementation.
- For optimization time, both Views and Rewrite incur higher planning overhead than Pure RLS. This is because they generate larger and more complex query trees, increasing the optimizer's search space and planning effort.

2. Black-Box Implementation

- In the black-box setting, cross query-policy optimization is disabled. The paper evaluates Pure RLS (implemented using SECURITY DEFINER functions), Indexed RLS, and Security Barrier Views.
- Policy-aware indexing significantly improves execution time and robustness for complex cyclic policies. In several cases, Indexed RLS successfully completes queries that time out under Pure RLS.
- Security Barrier Views generally introduce higher planning overhead and more variable execution behavior compared to Pure RLS, while Pure RLS provides a more stable performance profile.

Options to Explore

1. Policy Depth vs. Performance

How does policy dependency depth affect optimizer behavior, planning time, and execution time?

2. PostgreSQL vs. SQL Server RLS

A cross-engine study. Does SQL Server's RLS optimizer handle policy predicates more effectively than PostgreSQL under complex policy workloads?

3. Generalized Strategy Selection Algorithm

Develop a generalized algorithm that automatically determines whether to use Pure RLS, Views, or Query Rewrite based on query and policy characteristics. Can query and policy patterns be used to predict the most efficient enforcement strategy for a given workload?